Problem 1

1-(a) $h(n) = O(\log_2 n) : \exists (c_2>0, n_0>0)$ s.t. $h(n) \le c_2 \log_2 n = c_2 \log_2^3 \log_3 n$

let $c_2' = c_2 \log_2^3$, $n_0' = n_0$. Then (c_2', n_0') witnesses

$h(n) \le c_2' \log_3 n \Rightarrow h(n) = O(\log_2 n) = O(\log_3 n)$

1-(b) For asymptotically non-negative functions $f(n)$ and $g(n)$,

if $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0 \Rightarrow |\frac{f(n)}{g(n)} - 0| < \epsilon$, $0 \le \frac{f(n)}{g(n)} < \epsilon$, $f(n) = \epsilon g(n)$

let $c = \epsilon$ and $n_0 = n$, we have $f(n) < cg(n)\ \forall n \ge n_0$, which matches the definition of $f(n) = o g(n)$.

$\because \lim_{n\to\infty} \frac{f(n)}{g(n)} = \lim_{n\to\infty} (\frac{2^n}{3^n}) = 0 \quad \therefore f(n) = o g(n) \Rightarrow O(f(n)) \subsetneq O(g(n))$

Prove $f(n) = o g(n) \Rightarrow O(f(n)) \subsetneq O(g(n))$:

let $h(n) = O(f(n))$, By definition $\exists (c_1>0, n_0>0)$ s.t. $h(n) \le c_1 f(n)$

we known that $f(n) = o g(n)$

$\Rightarrow \exists (c>0, n_1>0)$ s.t. $f(n) < cg(n)\ \forall c>0, n \ge n_1$

Hence, choose $c=1$, we have $f(n) < g(n)$

Hence, we have $h(n) \le c_1 f(n) < c_1 g(n) \Rightarrow h(n) \in O(g(n))$

$\Rightarrow O(f(n)) \subseteq O(g(n))$

Next, we want to prove that $O(f(n)) \ne O(g(n))$:

since $g(n) \le 1 \cdot g(n) \Rightarrow g(n) \in O(g(n))$

assume $g(n) \in O(f(n)) \Rightarrow \exists (c_2>0, n_2>0)$ s.t. $g(n) \le c_2 f(n) \Rightarrow \frac{1}{c_2} g(n) \le f(n)$

we know $f(n) = o g(n)$, if we choose the constant to be $\frac{1}{2c_2}$, then we have: $f(n) < \frac{1}{2c_2} g(n) \Rightarrow g(n) \le c_2 f(n) < \frac{1}{2} g(n)$, since $g(n) \ge 0$, $\frac{1}{2} g(n)$ will never be larger than $g(n) \Rightarrow$ the initial assumption $g(n) \in O(g(n)) \in O(f(n))$ is false

$\Rightarrow$ it must be the case that $O(f(n)) \subsetneq O(g(n))$

1- (c) $n^n = \underbrace{n \times n \times \cdots \times n}_{n}$, $n! = \underbrace{1 \times 2 \times 3 \times \cdots \times n}_{n}$

$\forall n > 1$, $n^n > n! \Rightarrow \lim_{n\to\infty} \frac{g(n)}{f(n)} = \lim_{n\to\infty} \frac{n!}{n^n} = 0$

$\Rightarrow f(n) = \omega(g(n)) \Rightarrow g(n) = o(f(n)) \Rightarrow O(g(n)) \subseteq O(f(n))$

2. Consider $m \geq m_0$ and fixing m constant, $n \geq n_0$

$\exists (c_2 > 0, n_0 > 0, m_0 > 0)$ s.t. $mn \leq c_2 n \Rightarrow mn = O(n)$

$\exists (c_2' > 0, n_0' > 0, m_0' > 0)$ s.t. $m \log_2 n \leq c_2' \log_2 n \Rightarrow m \log_2 n = O(\log n)$

$\Rightarrow O(g(m,n)) \subseteq O(f(m,n))$

Consider $m \geq m_0$, $n \geq n_0$ and $m = n$,

$f(m,n) = 2n$, $g(m,n) = n \log_2 n$

$\exists (c_2 > 0, n_0 > 0)$ s.t. $2n \leq c_2 n \Rightarrow f(m,n) = O(n)$

$\exists (c_2' > 0, n_0' > 0)$ s.t. $n \log_2 n \leq c_2 n \log_2 n \Rightarrow g(m,n) = O(n \log n)$

$\Rightarrow O(f(m,n)) \subseteq O(g(m,n))$

Since $f(m,n)$ bounds $g(m,n)$ in some paths but not all, the relationship between them is none of the above.

3-(a)

		cost	reps
	DISCREET-SORT(arr)		
1	$i \leftarrow 0$	d_1	1
2	$n \leftarrow length(arr)$	d_2	1
3	**while** $i < n$	d_3	$n+1$
4	**if** $i = 0$ or $arr[i-1] \leq arr[i]$	d_4	n
5	$i \leftarrow i + 1$	d_5	n
6	**else**		
7	SWAP($arr[i], arr[i-1]$)	d_7	0
8	$i \leftarrow i - 1$	d_8	0

Best case: the array is in a sorted order

$T(n) = d_1 + d_2 + d_3(n+1) + d_4 n + d_5 n$ $\Rightarrow$ linear to n; $T(n) = \Theta(n)$

3-(b)

		cost	reps
	DISCREET-SORT(arr)		
1	$i \leftarrow 0$	d_1	1
2	$n \leftarrow length(arr)$	d_2	1
3	**while** $i < n$	d_3	
4	**if** $i = 0$ or $arr[i-1] \leq arr[i]$	d_4	r_4
5	$i \leftarrow i + 1$	d_5	r_5
6	**else**	d_6	0
7	SWAP($arr[i], arr[i-1]$)	d_7	r_7
8	$i \leftarrow i - 1$	d_8	r_8

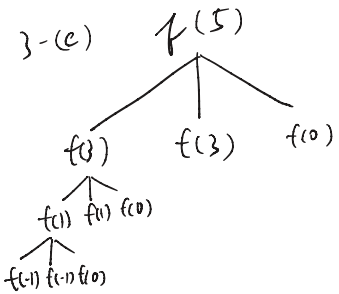
$r_4 = r_5 + r_8$, $r_5 = \sum_{\bar{i}=1}^{n-1} \bar{i} + n$, $r_7 = \sum_{\bar{i}=1}^{n-1} \bar{i}$, $r_8 = \sum_{\bar{i}=1}^{n-1} \bar{i}$

Worse case: the array is reversely sorted. Under this situation, when $\bar{i}$ moves one step forward, it will have to go all the way back to the front and back to its current place before making another step forward. The total backward steps is $\sum_{\bar{i}=1}^{n-1} \bar{i} = \frac{n(n-1)}{2}$, the total forward steps is $\sum_{\bar{i}=1}^{n-1} \bar{i} + n = \frac{n(n+1)}{2}$, and the total

steps for the while loop will be $\frac{n(n-1)}{2} + \frac{n(n+1)}{2} + 1$

Hence, the time complexity will be $\Theta(n^2)$.

3-(c)



for every number n,

when n is odd number, $n = 2k-1$, $k \in \mathbb{N}$, and k is the number of subtractions needed to make $n \le 0$. When n is even, $n = 2k$, $k \in \mathbb{N}$, and k is the number of subtractions needed to make $n \le 0$. k will be the total depth (layer) of the tree, and a tree will have $2^k = 2^{\frac{n}{2}}$ (or $2^{\frac{n+1}{2}}$ for odd numbers).

Hence, the overall time complexity will be $O(2^{\frac{n}{2}})$

worst-case

4. let $q = 1+x$, $q^n = (1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$, take the term where $k = p+1$: $q^n > \binom{n}{p+1} x^{p+1} = \frac{n(n-1)\cdots(n-p)}{(p+1)!} x^{p+1}$

let $c = \frac{x^{p+1}}{(p+1)!}$ and $n(n-1)\cdots(n-p) = n^{p+1} + c_p n^p + c_{p-1} n^{p-1} + \cdots + p > n^p$

$q^n > c\,(n^{p+1} + c_p n^p + \cdots + p) = cn^{p+1} + c \cdot c_p n^p + \cdots + cp > c \cdot c_p n^p$

$n^p < \frac{1}{c \cdot c_p} q^n$, let $c_2 = \frac{1}{c \cdot c_p}$, $\exists (c_2 > 0, n_0 > 0)$ s.t. $n^p \le c_2 q^n$

$\forall n \ge n_0 \Rightarrow n^p = O(q^n)$